

CYBER SECURITY COURSE

File Carving and Deleted Data Recovery

Professional Case Report 06

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Report date	24. septembar 2026
Case ID	CSC-2026-06
Discipline	Digital forensics
Overall risk	Medium
Evidence status	Technique demonstrated in saved screenshots

Purpose

Locate file signatures in hexadecimal data, define the correct byte range and export a recovered object without modifying the source evidence.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A raw evidence image contained recoverable content that was not available through normal file-system navigation. The exercise demonstrated that deletion does not equal destruction. A defensible carving process depends on read-only handling, precise offsets and hash verification.

Case Profile

Scenario	A raw evidence image contained recoverable content that was not available through normal file-system navigation.
Environment	Forensic image examined in WinHex using read-only navigation and block selection.
Tools	WinHex, Hexadecimal analysis, File signatures, Read-only evidence handling

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Opened the evidence image without writing to the source.
2. Located candidate headers and validated surrounding byte structure.
3. Selected a block from file header to the corresponding footer or calculated length.
4. Exported the selected block and validated the recovered file type.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Content remains recoverable	Medium	Deleted or unallocated data can persist until its storage blocks are overwritten.
F02	Signature-based recovery	Medium	File headers and footers allow recovery when directory metadata is missing.
F03	Evidence integrity	High	Recovery work must operate on a copy and document offsets to preserve reproducibility.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
Identify magic bytes
Mark beginning and end of block
Save selected block to a new evidence file
```

Risk Assessment

Risk	Level	Operational impact
Data exposure	Medium	Deleted sensitive files may still be recoverable
Forensic contamination	High	Writing to original media can invalidate evidence

Recommendations

1. Use write blockers or verified forensic copies.
2. Record source hash, offsets and output hash for every carved object.
3. Validate recovered files with both signatures and an independent viewer.
4. Securely sanitize retired media when recovery is not required.

Knowledge Demonstrated

- Hex analysis
- File signature recognition
- Evidence preservation
- Recovery validation

Conclusion

The exercise demonstrated that deletion does not equal destruction. A defensible carving process depends on read-only handling, precise offsets and hash verification.

Hands On Lab Practical 06

Completed as part of the authorized cyber security course practical work.